

Travel Bokka

**AGENTRIX 2026
Interim Submission**

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1. Chosen Domain

Travel Technology & Intelligent Transportation Systems

Our solution utilizes AI-powered travel technology to address the needs of solo and budget independent travelers in Sri Lanka, focusing on real-time decision-making, multimodal guidance, and low-connectivity resilience.

2. Problem Statement

Independent travelers in Sri Lanka face severe logistical and financial challenges due to a fragmented information infrastructure. We have identified four critical failure points in the current travel ecosystem:

- **Transit Volatility & Collapse:** Local public transit operates with zero unified coordination. Unannounced cancellations, strikes, or weather disruptions frequently cause complete itinerary collapses, leaving tourists stranded without viable, real-time recovery paths.
- **The "Tourist Tax" & Scam Vulnerability:** Information asymmetry and distinct language barriers restrict fair negotiation within the local informal economy. Tourists are routinely subjected to "fake closure" scams, price gouging by unregulated transport (e.g- tuk-tuks), and predatory up-selling.
- **Static Information Decay:** Travelers are forced to rely on outdated, static travel blogs or fragmented forums, lacking real-time, context-aware routing that accounts for changing local conditions.
- **Connectivity Blackouts:** Frequent internet connectivity dropouts in rural, mountainous, and scenic areas (e.g., Ella, Sigiriya) render conventional, cloud-dependent navigation and translation applications completely unusable exactly when travelers need them the most.

3. Solution Outline

We are developing **Travel Bokka**, an Autonomous Omni-Channel Tourist Guardian. It is an offline-first mobile application that acts as a pocket-sized local expert, providing travelers with real-time route recovery, cultural guidance, and proactive scam protection.

Core Features:

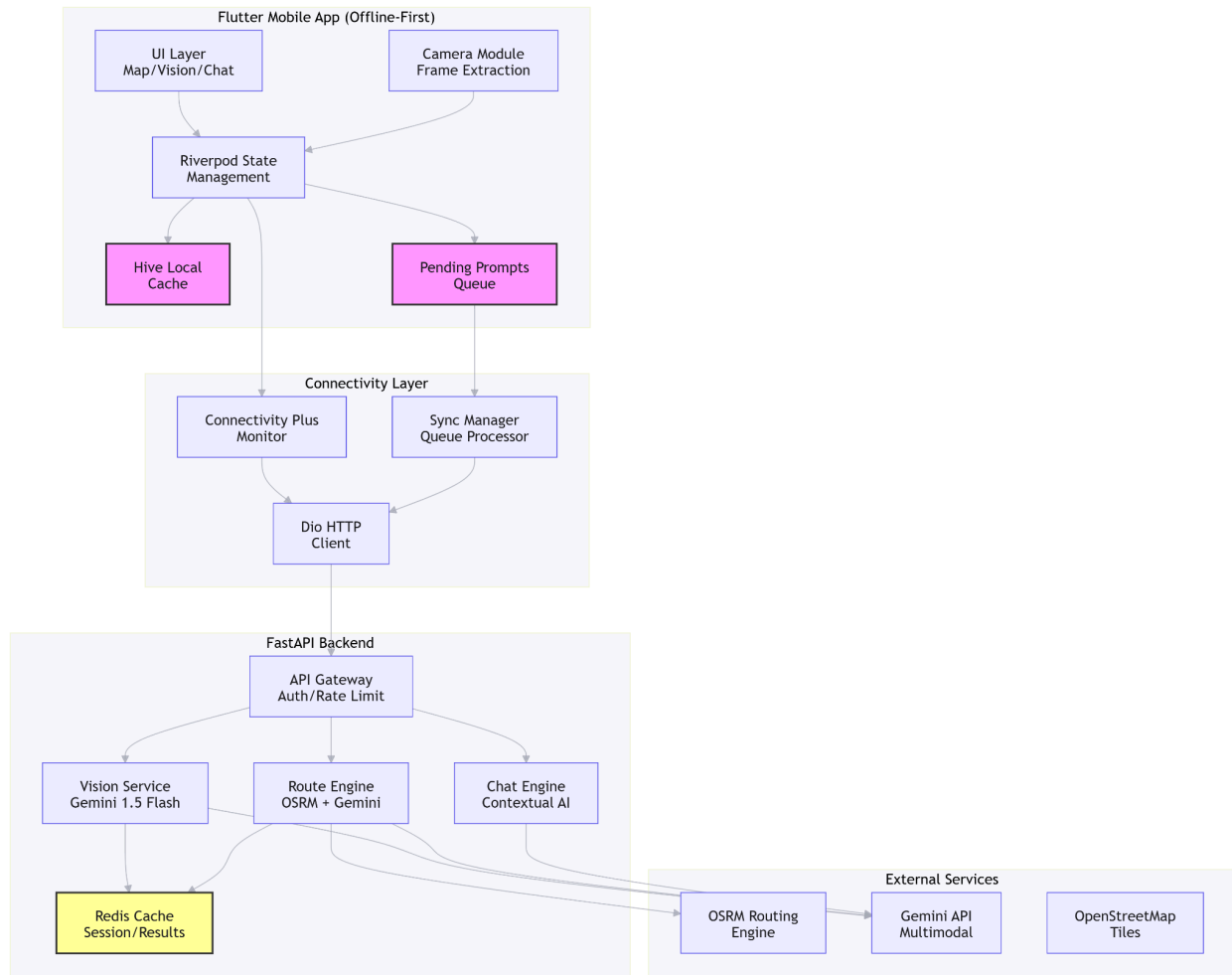
- **Reactive Route Pivot Engine:** Actively recalculates alternative paths upon transit disruptions. Unlike standard maps, it enriches the new route with localized linguistic scripts (e.g., how to ask for the correct bus in Sinhala) and exact expected fare costs to prevent exploitation during the detour.
- **The "Guardian" Conversation Auditor:** A background audio-processing engine that listens to live negotiations (e.g., with a tuk-tuk driver). It cross-references spoken claims against a database of local truths, instantly triggering a visual UI alert if the traveler is being lied to or overcharged.
- **Sight-Glass Multimodal Guide:** Allows users to scan physical environments. Whether pointing the camera at a historical monument for architectural context or scanning a local restaurant menu to detect hidden "tourist markups," the app processes visual data in real-time.
- **Offline-First "Survival Packs":** Utilizes local caching (Hive) to download hyper-localized JSON RAG state data. If a user enters a zero-connectivity zone, core functionalities (routing geometry, cultural rules, cached vision results, and emergency contacts) remain fully active.

4. How We Plan to Apply AI

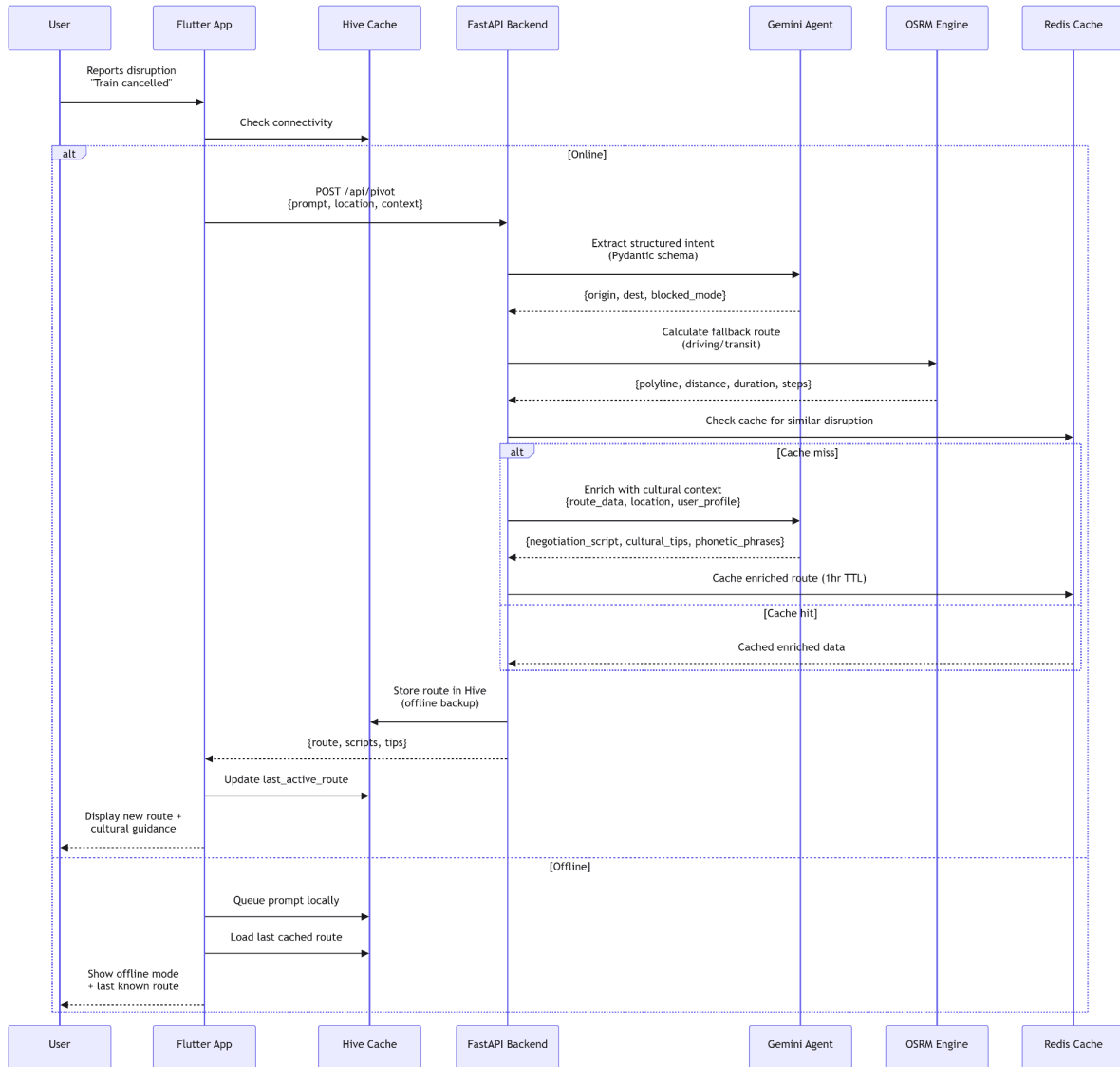
Our system employs a highly resilient **Hybrid Architecture**, combining deterministic APIs with Agentic LLM orchestration to ensure absolute reliability, mathematical accuracy, and dynamic context-awareness.

- **Multimodal Vision (Gemini 1.5 Flash/Pro):** We utilize Gemini's native vision capabilities to process user-captured frames. To guarantee application stability, we enforce strict **Pydantic structured JSON outputs**, ensuring the LLM extracts exact pricing data and cultural context without breaking the Flutter frontend via hallucinations.
- **Agentic Orchestration (LangChain + CrewAI):** We deploy a multi-agent workflow. A 'Route Crew' translates natural language disruption alerts into actionable routing parameters. Simultaneously, a single-agent 'Guardian' uses tool-calling to audit real-time audio transcripts for deceptive tactics.
- **Retrieval-Augmented Generation (RAG):** We utilize Supabase (pgvector) and local FAISS vector stores to anchor the LLMs to a local ground truth. The AI continuously searches a specialized knowledge base of average transport fares, official opening hours, and verified scam playbooks, preventing the LLM from generating plausible but incorrect local advice.
- **Hybrid Routing Integration:** We explicitly separate "math" from "reasoning." The AI acts as the orchestrator but defers to deterministic graph routing (OSRM Public API / Google Maps API) for exact distances and geometry. The AI then wraps this geometry in rich, contextual advice, preventing the dangerous LLM hallucination of physical travel logistics.

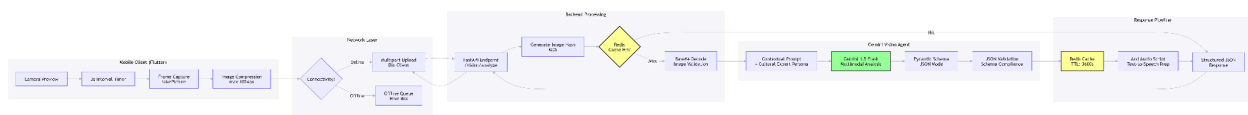
1. Overall System Architecture



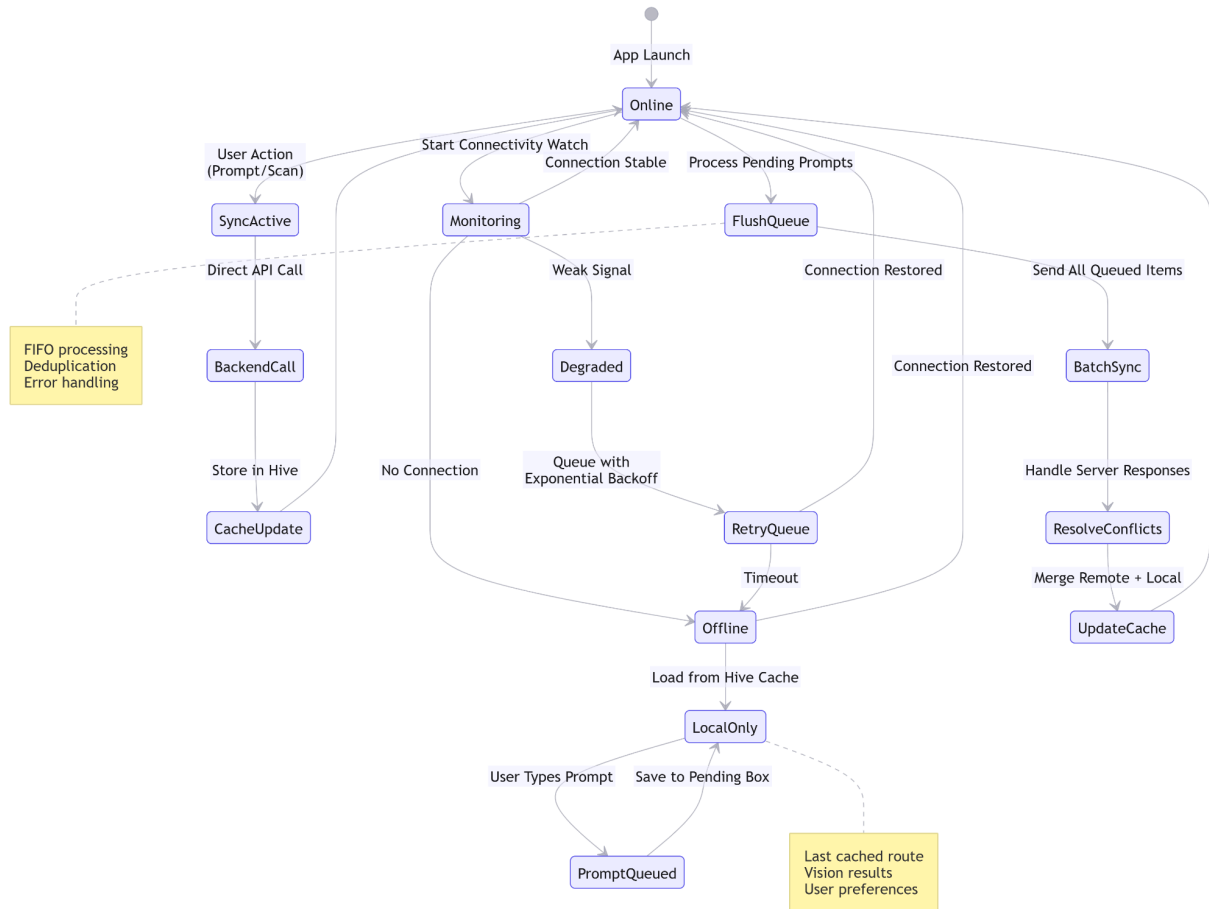
2. Agentic Workflow: Reactive Route Pivot Engine



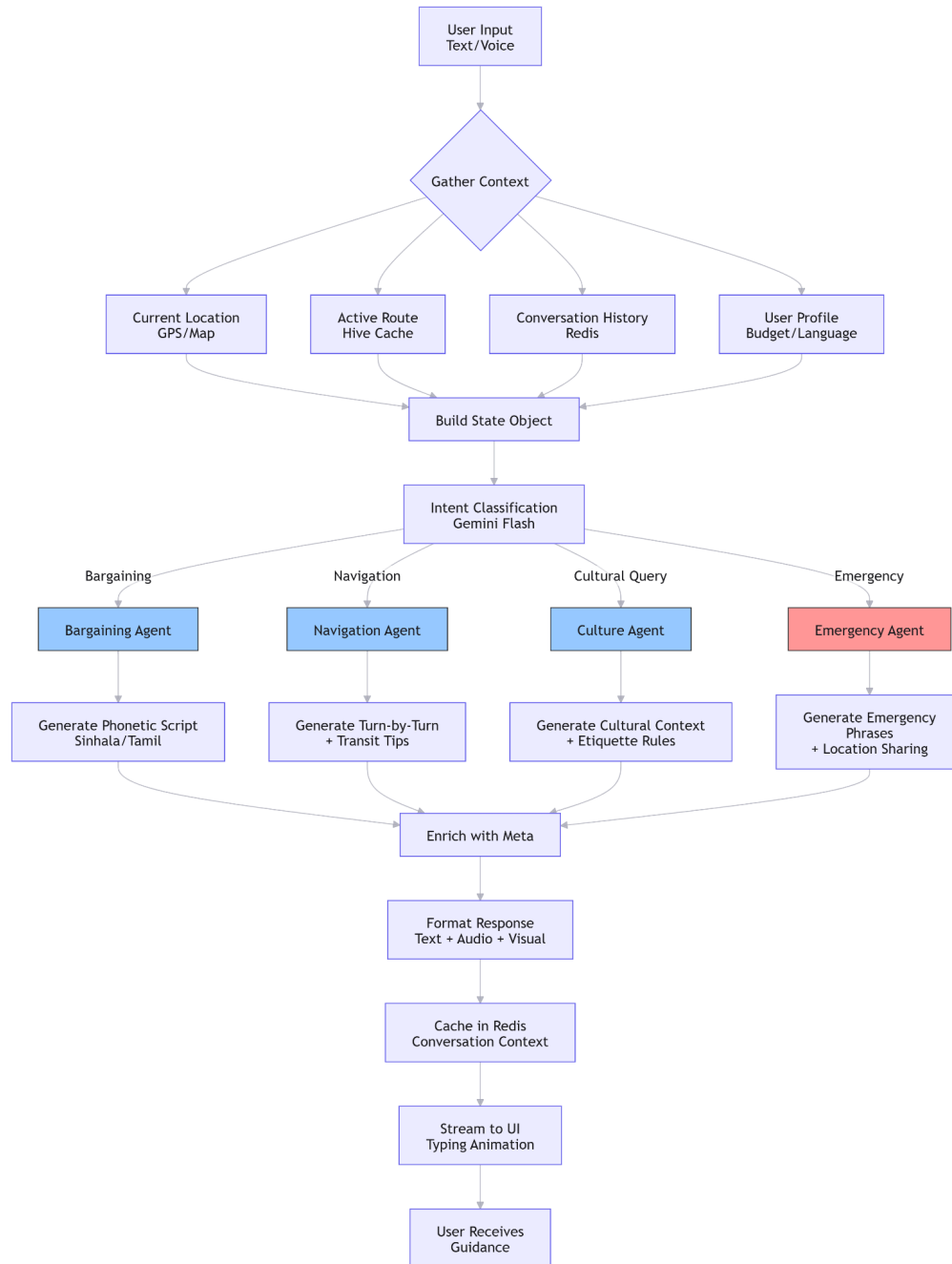
3. Multimodal Vision Processing Pipeline



4. Offline-First Data Synchronization Flow



5. Contextual Conversational Agent Workflow



6. End-to-End User Journey: Colombo to Ella Disruption

